

Superstore Sales Analysis

1. Project overview

This project analyzes Global Superstore's sales records of 51,000 rows from 2011-2014. The goal is to uncover sales trends, purchasing behaviour of customers, which countries buy the most, which product cause losses, and how discounts and shipping choices affect the total profit.

2. Dataset summary

- Rows – 51,290 rows
- Columns – 27 columns
- Key features
 - Global presence - Present across 147 different countries
 - Product variety - Features 3,788 unique products across diverse categories (e.g., Technology, Furniture)
 - Regional markets – Strongly dominates major markets including APAC, LATAM, and EU

3. Data Exploration using Excel

We begin with Data preparation and cleaning

- **Data loading:** loaded the csv file into excel.
- **Data Cleaning:** removed unnecessary and unwanted columns that had no Impact on this project.
- **Column standardization:** renamed column names to snake casing for better readability and documentation.
- **Data Types:** changed data types of some columns like order date and shipping date to DATE format.
- **Data processing:** prepared the data for Analysis and moved to MySQL workbench.

4. Data Analysis using SQL

After initial data exploration and data cleaning, we load the cleaned data into Mysql workbench for structured data analysis and answer key business questions.

- *Revenue by Category:* Compared total revenue across each category

	category	total_sales
▶	Technology	4744691
	Furniture	4110884
	Office Supplies	3787330

- *Top 5 countries with highest Revenue:* 42% of the total revenue generated comes from the Top 5 countries. Among those countries, United States alone contributes 18.17% of the Total Revenue generated.

	country	country_sales	sales_percentage
▶	United States	2297354	18.17
	Australia	925257	7.32
	France	858930	6.79
	China	700591	5.54
	Germany	628857	4.97

- *Products vs revenue:* "Found that the top-performing products by revenue all fall under the Technology category"

	product_name	category	total_sales	order_count
▶	Apple Smart Phone, Full Size	Technology	86936	51
	Cisco Smart Phone, Full Size	Technology	76441	38
	Motorola Smart Phone, Full Size	Technology	73159	38
	Nokia Smart Phone, Full Size	Technology	71904	47
	Canon imageCLASS 2200 Advanced Copier	Technology	61600	5

- *Loss making Products:* Found out the worst performing Products.

	product_name	profit
▶	Cubify CubeX 3D Printer Double Head Print	-6599.98
	Motorola Smart Phone, Cordless	-4088.38
	Cubify CubeX 3D Printer Triple Head Print	-3839.99
	GBC DocuBind P400 Electric Binding System	-3701.89
	Lexmark MX611dhe Monochrome Laser Printer	-3399.98
	Hoover Stove, White	-3059.82
	Apple Smart Phone, Full Size	-3009.44
	Ibico EPK-21 Electric Binding System	-2929.48
	Barricks Conference Table, Rectangular	-2750.28
	Cubify CubeX 3D Printer Double Head Print	-2639.99

- *Revenue by Segment:* compared revenue across different Segments. "The **Consumer** segment is the largest driver of top-line revenue, followed by Corporate and Home Office"

	segment	total_sales
▶	Consumer	6508141
	Corporate	3824808
	Home Office	2309956

- *Discount vs Revenue & Profit:* Categorized discount into "no discount, low discount, medium and high discount" for easy interpretation of the discount column. Found out that high discount leads to negative profits.

	discount_category	total_sales	avg_profit
►	No Discount	6992734	61.04
	Low Discount	3824205	45.72
	Medium Discount	1454342	-58.16
	High Discount	371624	-98.89

- **Shipping analysis:** "Found an inverse correlation: shorter shipping durations correspond to higher average shipping costs"

	ship_mode	avg_shipping_cost	avg_shipping_duration
►	Same Day	42.94	0days 5hours 54minutes
	First Class	41.05	2days 21hours 22minutes
	Second Class	30.47	3days 19hours 31minutes
	Standard Class	19.97	4days 21hours 57minutes

- **Yearly sales Trends:** compared sales in each year from 2011 – 2014. Found out that the revenue increased from 17.87% in 2011 to 34.01% in 2014 representing a whopping 90% increase in just 3 years

	year	total_sales	previous_sales	next_sales	sales_percentage	yoy_percentage_increase
►	2011	2259511	NULL	2677493	17.87	NULL
	2012	2677493	2259511	3405860	21.18	18.50
	2013	3405860	2677493	4300041	26.94	27.20
	2014	4300041	3405860	NULL	34.01	26.25

- **Quarterly sales Analysis:** created a month quarter column to get a snapshot of short-term business growth and found that Q3 & Q4 are the best performing Quarters.

	month_quarter	total_sales	quarterly_percentage_increase
►	Q1	1989428	NULL
	Q2	2872415	44.38
	Q3	3480707	21.18
	Q4	4300355	23.55

- **Monthly Sales Trends:** compared how business is performing across months in different years. Found out that December is present in top 3 in every year. November and September came 3 times among top 3 in 4years. August and June are sales anomalies or there might be bulk orders or “go-to-school” shopping.

	year	total_sales	top_3_months
►	2011	2259511	December, November, September
	2012	2677493	December, November, August
	2013	3405860	December, June, September
	2014	4300041	November, December, September

5. Business Recommendations:

- **VIP Customer Programs:** Launch localized loyalty programs in the top 5 countries to increase repeat purchase rates, as retaining these customers is far cheaper than acquiring new ones elsewhere.

- **Aggressive Q4 Ad Spend:** Allocate 60% to 70% of your annual marketing and advertising budget strictly to October, November, and December, where consumer buying intent is naturally at its absolute peak.
- **Targeted Loyalty Rewards:** Reserve medium/high discounts exclusively for dead stock (inventory not moving) or your highest-lifetime-value customers, rather than offering them universally.